Hsuan-Fu (Jeff) Hsu

AI and ML Engineering | Retrieval, RAG and Model Evaluation

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EDUCATION

University of Southern California — Los Angeles, CA
Master of Science in Computer Science | GPA: 3.6/4.0 — Jan 2026 - May 2027 (Expected)

National Taipei University — New Taipei City, Taiwan
Bachelor of Science in Computer Science | GPA: 3.5/4.0 — Sept 2020 - June 2024

SKILLS

Proficient: Python, C++, C#, Git

Familiar: PyTorch, Sentence Transformers, FAISS, FastAPI, NumPy, Docker, pytest, GitHub Actions, AWS, SQL, TypeScript

Exposure: TensorFlow, tsfresh, Ollama, OpenAI Agents SDK, LangChain, LangGraph, Google Cloud Run, Vitest

EXPERIENCE

National Taipei University — New Taipei City, Taiwan
Undergraduate Researcher | Advisor: Prof. Bor-Shing Lin — July 2022 - Feb 2024

- Built the **Python** signal pipeline for 30 participants at 2 wavelengths and compared 5 ML and 4 DL models; the team's best model reached 10.3% MAPE under leave-one-subject-out testing.
- Won individual NSTC research funding; 4th place in NTPU's project competition (team of 3).

Tech-Management Workshop Co., Ltd. — Taichung, Taiwan
Software Engineer Intern — July 2026 - Aug 2026

- Built a government birth-grant app on my own with **ASP.NET Core MVC** and **SQL Server**, deployed to 29 offices.
- Tuned SQL Server processing and storage, cutting database latency by at least 50% and resource use by at least 30%.

PROJECTS

Scientific Claim QA: RAG Evaluation — Oct 2026
Python, FastAPI, FAISS, Ollama, Docker | GitHub

- Built a **FastAPI** RAG service (FAISS, local LLM via Ollama) and tested 6 setups on 300 claims: a reranker raised nDCG@10 from 0.645 to 0.695, but faithful answers stayed at 79–84% in all 6.
- Dropped an LLM judge that matched 20 hand labels no better than chance (kappa −0.07); shipped with API keys, rate limits, **Docker**, and 112 tests in CI.

Scientific Retrieval Fine-tuning — Personal Fork — Oct 2026
Python, PyTorch, Sentence Transformers | GitHub

- Fine-tuned **MiniLM** with **PyTorch** on mined hard negatives: +3.2% nDCG@10 and +2.1% Recall@10 over random negatives on 300 SciFact test claims, 3 seeds.
- Added data leakage checks, a false-negative review, and 53 unit tests.

Promptfoo Agent Retry Validation — Personal Fork — Oct 2026
TypeScript, Vitest, OpenAI Agents SDK | GitHub

- Added nested error paths to retry-policy checks in **TypeScript**; 6 regression cases fail before the change and pass after.
- Wrote 81 **Vitest** cases, including 60 config checks and 11 retry scenarios with fake model calls.

Classroom AC Setpoint Simulator — USC IEEE HackIoT (team of 4) — Oct 2026
TypeScript, Next.js, Raspberry Pi 4, YOLO | GitHub

- Ported a teammate's Python AC model to **TypeScript** (matched over 1,344 hours) and built a Next.js simulator; one room used an estimated 27% less energy in an 8-week simulation.
- Set up a **Raspberry Pi 4** with teammates to count people with YOLO at about 7 FPS and read a DHT11 sensor twice a minute, sending data to Supabase.

PUBLICATION

Coauthor, "Noninvasive Portable System With Dual Near-Infrared Light Technology for Monitoring Uric Acid Concentration," IEEE Transactions on Human-Machine Systems, accepted for publication, Aug 2026.